Comments

The Comments section is for short papers which comment on papers previously published in **The Physical Review.** *Manuscripts intended for this section must be accompanied by a brief abstract for information retrieval purposes and a keyword abstract.*

Comment on "Truncated space calculation of 'exact' bound state energies"

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The recently suggested method of Bassichis and Strayer is compared with that of Révai and generalized for solving the A-body Schrödinger equation with $A > 1$.

[NUCLEAR STRUCTURE Schrödinger equation, separable potential, effective interaction in model space.]

In their recent paper, Bassichis and Strayer (BS)[1] suggested how to determine the bound state energies of the one-body Hamiltonian $H=T+PVP$, $P=\sum_{\alpha=1}^{N}|\alpha\rangle\langle\alpha|$, where $|\alpha\rangle$ are oscillator wave functions and T denotes the kinetic energy operator. It is our present aim to add two remarks:

(1) The method is based on separability of the potential. Using the algebraic (matrix) identity

$$P\frac{1}{T-\epsilon}P=P\frac{1}{PTQ[1/(QTQ-\epsilon)]QTP-PTP+\epsilon}P \tag{1}$$

($Q=1-P$) and the Schrödinger equation written in the form

$$|\epsilon\rangle=\frac{1}{T-\epsilon}PVP|\epsilon\rangle\,, \tag{2}$$

we arrive at two different (equivalent, of course) model space equations

$$\begin{aligned} P\frac{1}{P[1/(T-\epsilon)]P}P|\epsilon\rangle &\underset{(\mathrm{R})}{=} PVP|\epsilon\rangle \\ &\underset{(\mathrm{BS})}{=} PTQ\frac{1}{QTQ-\epsilon}QTP|\epsilon\rangle \\ &\quad -P(T-\epsilon)P|\epsilon\rangle \end{aligned} \tag{3}$$

showing the relation of the BS method to that of Révai.[2] Hence, the existence of a weakly bound state should be investigated carefully due to the possible oscillations (nonvariational character) of $\epsilon=\epsilon(N)$ for $N\rightarrow\infty$. Fortunately, they may simply be eliminated—for details see Ref. 2.

(2) In Révai's formulation, the transition to A-body interpretation of H and $|\alpha\rangle$ with $A>1$, i.e., the possibility of evaluating (=integrating) $P(T-\epsilon)^{-1}P$, is hindered by the auxiliary[3] factorization

$$\begin{aligned}&(k_1^2+\cdots+k_A^2-\epsilon)^{-1}\\ &\qquad=\text{const}\times\int_a^b ds\,\exp s(k_1^2+\cdots+k_A^2-\epsilon)\,.\end{aligned}$$

In the BS continued fraction approach, it is possible to avoid the numerical integrations entirely. For $A>1$, the corresponding matrix generalization of the continued fraction expansion of the right-hand side of Eq. (3) was described in Ref. 4. In the Brueckner theory context, it was also numerically tested[4] for $A=2$ and the convergence proved to be as quick as for $A=1$.

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[1] W. H. Bassichis and M. R. Strayer, Phys. Rev. C 18, 1505 (1978).
[2] J. Révai, in *Proceedings of the International Symposium on Nuclear Reactions, Balatonfüred, 1977,* edited by L. P. Csernai (Central Research Institute for Physics, Budapest, 1978), p. 390.
[3] E. Truhlík, private commmunication.
[4] M. Znojil, J. Phys. A: Math. Gen. 11, 1501 (1978).

